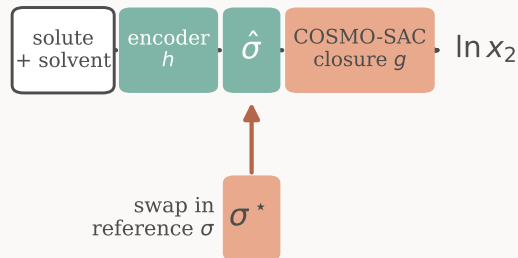


When do reference physical inputs help a learned solubility model?

1. The paradox

fixed, no fitted parameters



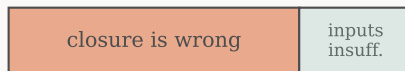
Premise: truer physical inputs → better.

**Here, the opposite. So which is wrong:
the closure, or the inputs?**

2. The measurement

*the reference input has an external oracle (VT-2005),
so the reference-input error splits exactly:*

$$B = B_{\text{closure}} + B_{\text{insuff}}$$



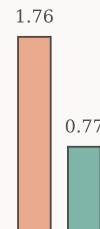
$$B_{\text{closure}} > B_{\text{insuff}} \quad (P_{\text{boot}} \approx 0.78)$$

**The ceiling is the CLOSURE,
not the inputs.**

3. No lever found

The obvious fix, a better closure,
does not hold up on representative data.

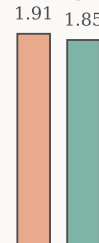
+56%,
grazes 0



2002 2010
(no dsp)

curated
n = 60

+3%,
not sig.



2002 2010/
dsp

represent.
n = 477

No closure-fidelity lever established.

What the paper establishes, graded by evidence

CERTIFIED

- Reference σ hurts solubility (3 seeds, $n = 5608$)
- Exact split $B = B_{\text{closure}} + B_{\text{insuff}}$ (Lemma 2)
- pKa flip: helps meta/para, hurts ortho ($\approx 20/20$)

LIKELY

- Latent: low-rank transferable surrogate ($n = 44$)
- Closure is the larger term (low- γ , $P_{\text{boot}} \approx 0.78$)
- Attribution holds on the matched $n = 60$ corner

OPEN

- No fidelity lever: 2010/dsp +3% ($n = 477$, $P = 0.62$)
- Transfer of the attribution to finite composition
- Whether the drift is the closure's inverse $J^+(m-g)$